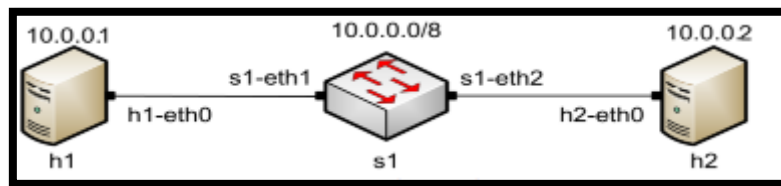


Miniedit

In this lab, you will use the application MiniEdit to deploy the topology illustrated below.



What is Miniedit?

MiniEdit is a simple GUI network editor for Mininet. MiniEdit is an experimental tool created to demonstrate how Mininet can be extended.

How to run Miniedit?

MiniEdit can be started by using any of the following method depending on your way of installation of MiniNet.

- **For Pre-Built MiniNet VM:**

Use SSH command to connect to the machine, as described in the previous lab. In that way, the host machines graphics card will be used to turn on the MineEdit GUI.

- **For Mininet installation using source files:**

If you have installed your Mininet in Ubuntu and used the source files to install Mininet. You can run the following command to open the MiniEdit GUI. In this case, the graphics of your host machine Ubuntu will be used to turn on the MiniEdit.

Go to the **/mininet/mininet/examples** folder and run this command

```
Python3 miniedit.py
```

Command syntax to start Miniedit.

```
Python3 /path/to/file/miniedit.py
```

If you are using python 2 then use the python appropriately. The new versions are all using python3.



The screenshot shows the MiniEdit software interface. At the top is a menu bar with 'File', 'Edit', 'Run', and 'Help'. On the left is a vertical toolbar containing several icons. Six numbered callouts point to specific elements:

- (1) Select: Points to the first icon in the toolbar, which is a mouse cursor.
- (2) Host: Points to the second icon in the toolbar, which is a black square.
- (3) Legacy switch: Points to the third icon in the toolbar, which is a blue circular arrow.
- (4) Link: Points to the fourth icon in the toolbar, which is a blue square with a white 'X'.
- (5) Run: Points to the 'Run' button at the bottom of the toolbar, which is a green button with the word 'Run' in white.
- (6) Stop: Points to the 'Stop' button at the bottom of the toolbar, which is a red button with the word 'Stop' in white.

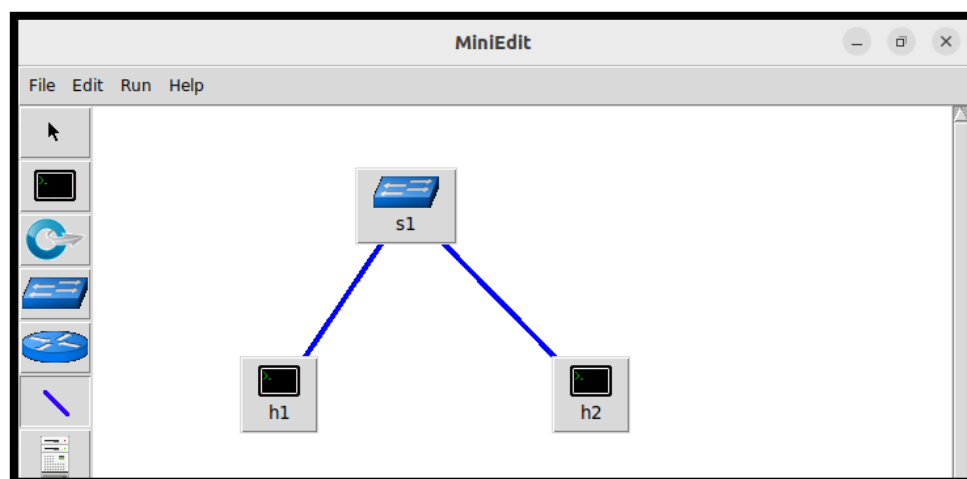
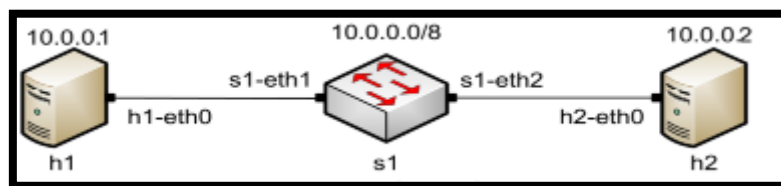
The buttons are:

1. **Select:** allows selection/movement of the devices. Pressing Del on the keyboard after selecting the device removes it from the topology.

2. **Host:** allows addition of a new host to the topology. After clicking this button, click anywhere in the blank canvas to insert a new host.
3. **Legacy switch:** allows addition of a new legacy switch to the topology. After clicking this button, click anywhere in the blank canvas to insert the switch.
4. **Link:** connects devices in the topology (mainly switches and hosts). After clicking this button, click on a device and drag to the second device to which the link is to be established.
5. **Run:** starts the emulation. After designing and configuring the topology, click the run button.
6. **Stop:** stops the emulation.

Building topology in MiniEdit:

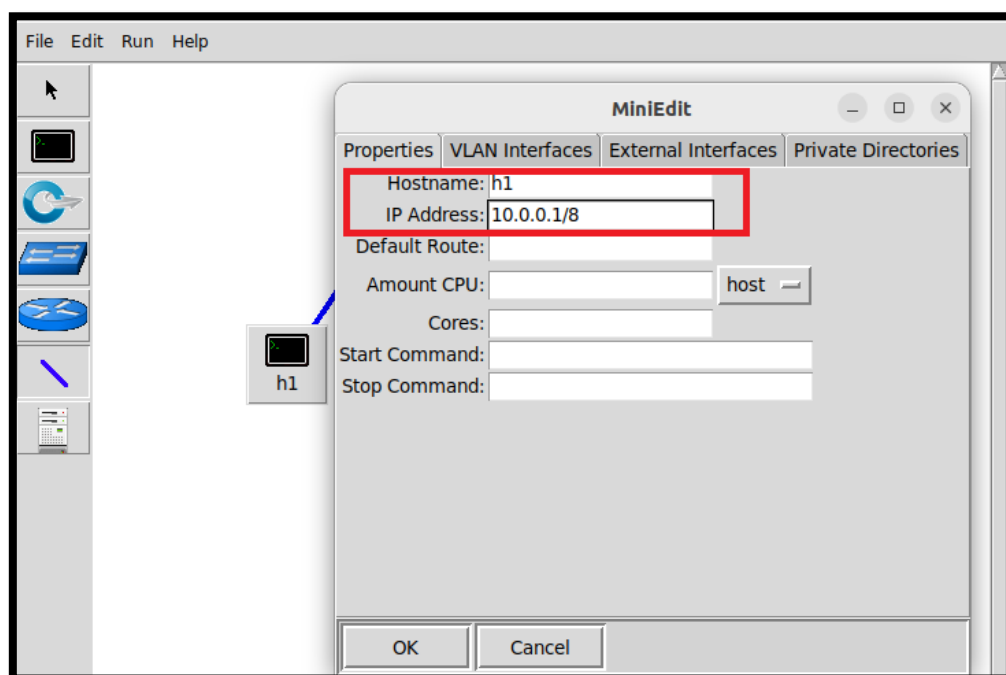
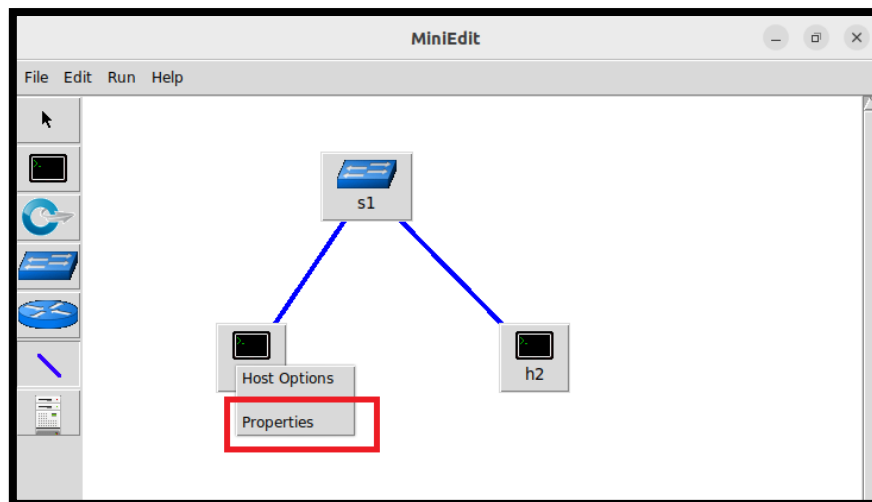
Create following topology in the MinEdit.



Use the buttons described in the previous step to add and connect devices. The configuration of IP addresses is described below.

IP configuration:

Configure the IP addresses of host h1 and host h2. Host h1's IP address is 10.0.0.1/8 and host h2's IP address is 10.0.0.2/8. A host can be configured by holding the right click and selecting properties on the device. For example, host h2 is assigned the IP address 10.0.0.2/8 in the figure below. Click OK for the settings to be applied.

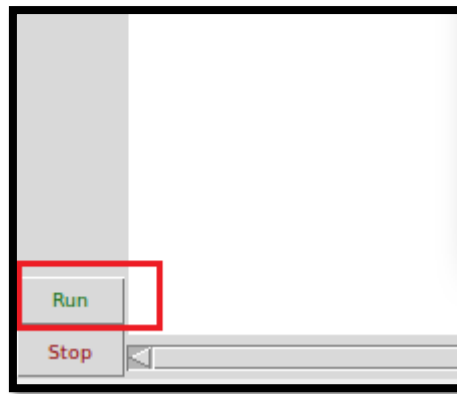


In a similar way, assign IP of 10.0.0.2 to h2 PC/server. You do not need to assign IP to a switch here.

Test connectivity:

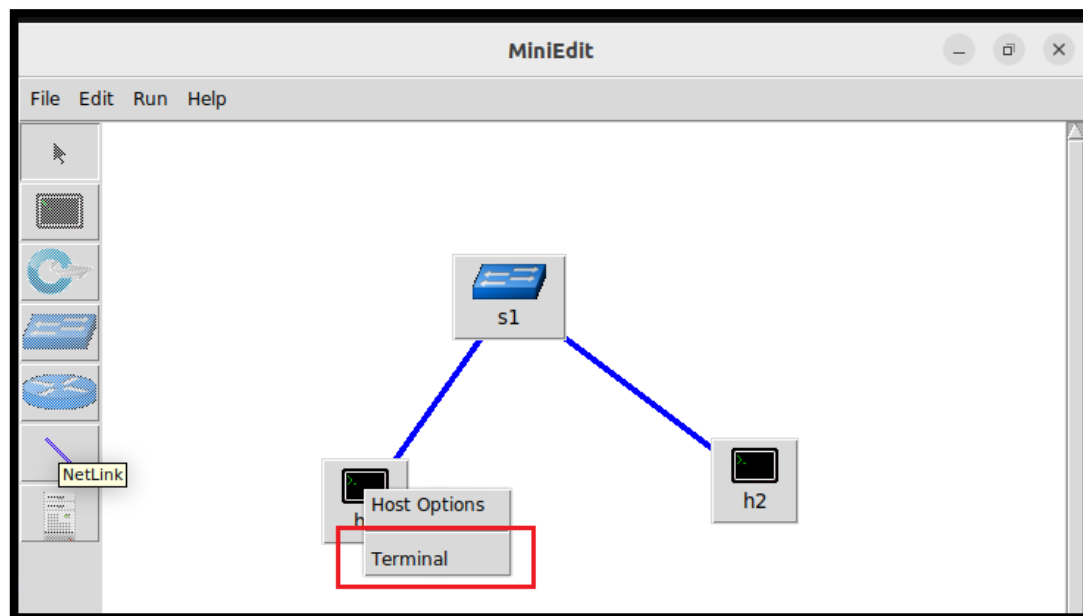
Before testing the connection between host h1 and host h2, the emulation must be started.

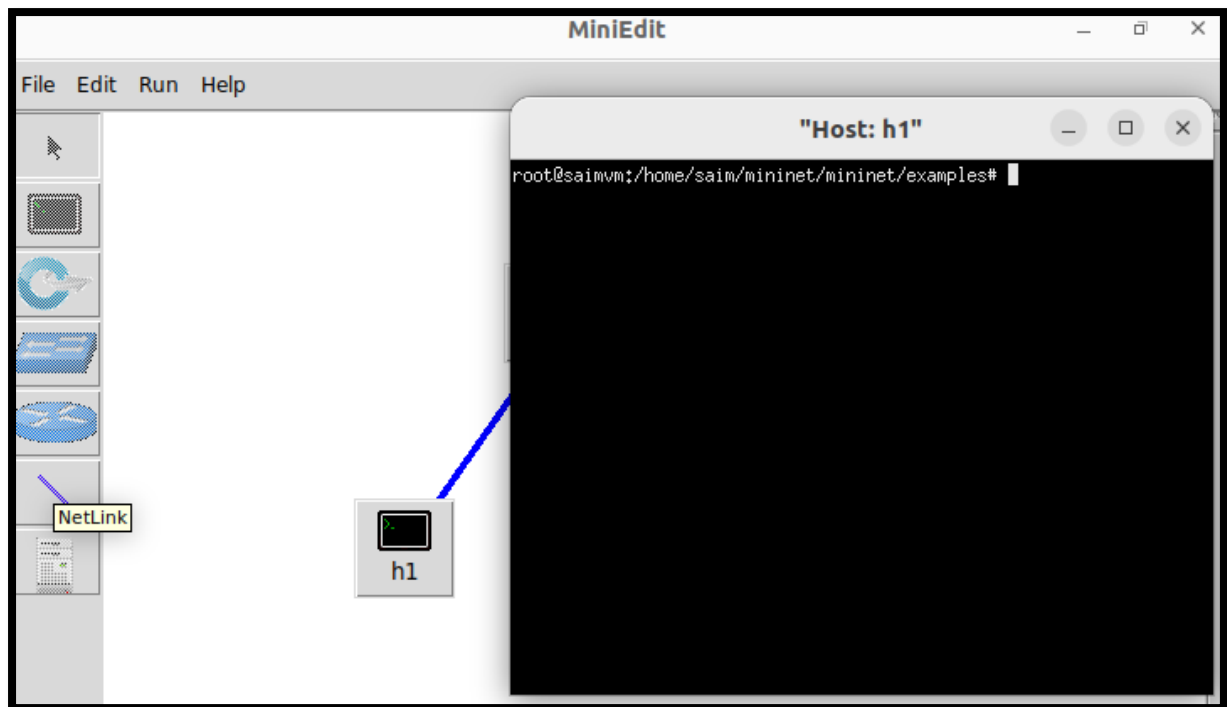
Click on the Run button to start the emulation. The emulation will start and the buttons of the MiniEdit panel will gray out, indicating that they are currently disabled.



The network and terminals at host h1 and host h2 will be available for testing.

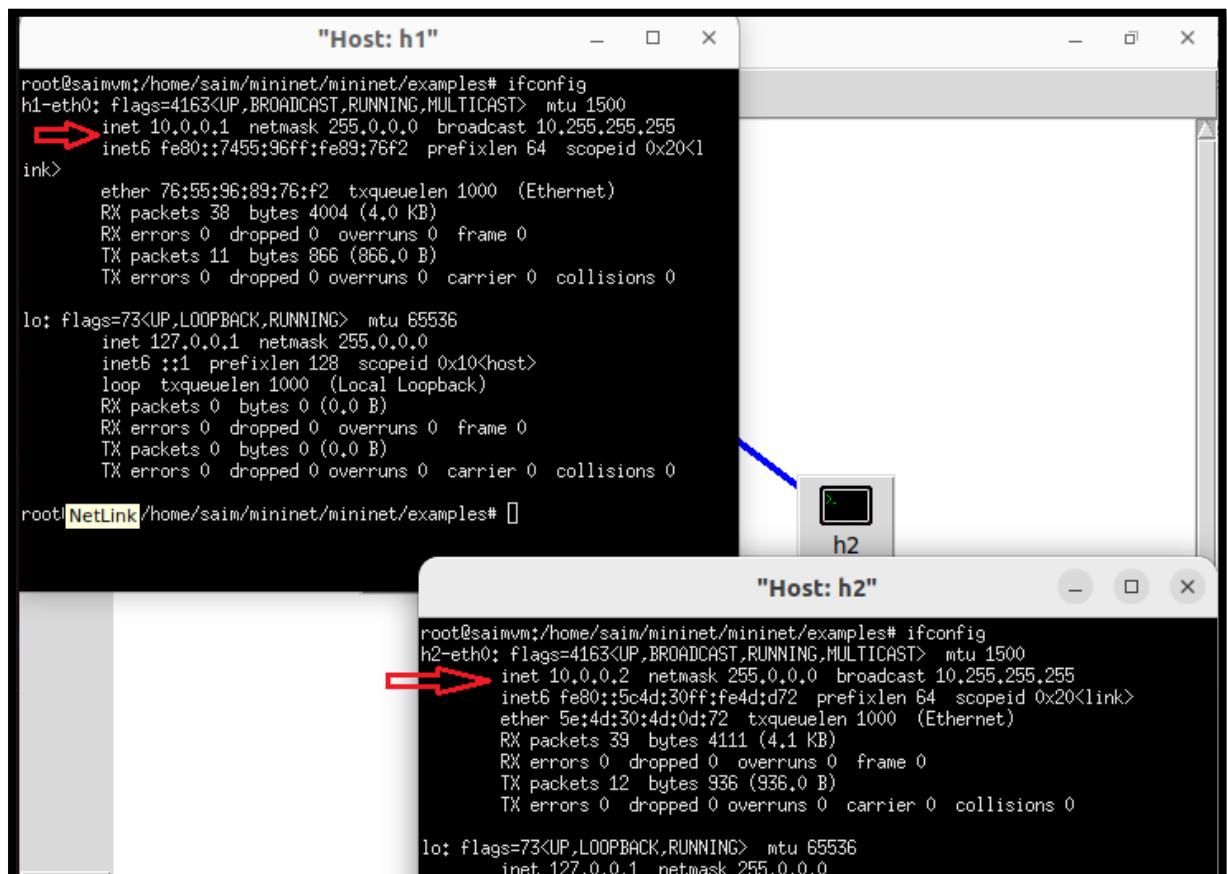
Now, right click on the host machine and open the Terminal for each host to test the connectivity. As shown in the figure below.





On host h1's terminal, type the command shown below to display its assigned IP addresses. The interface h1-eth0 at host h1 should be configured with the IP address 10.0.0.1 and subnet mask 255.0.0.0. Do same for the h2.

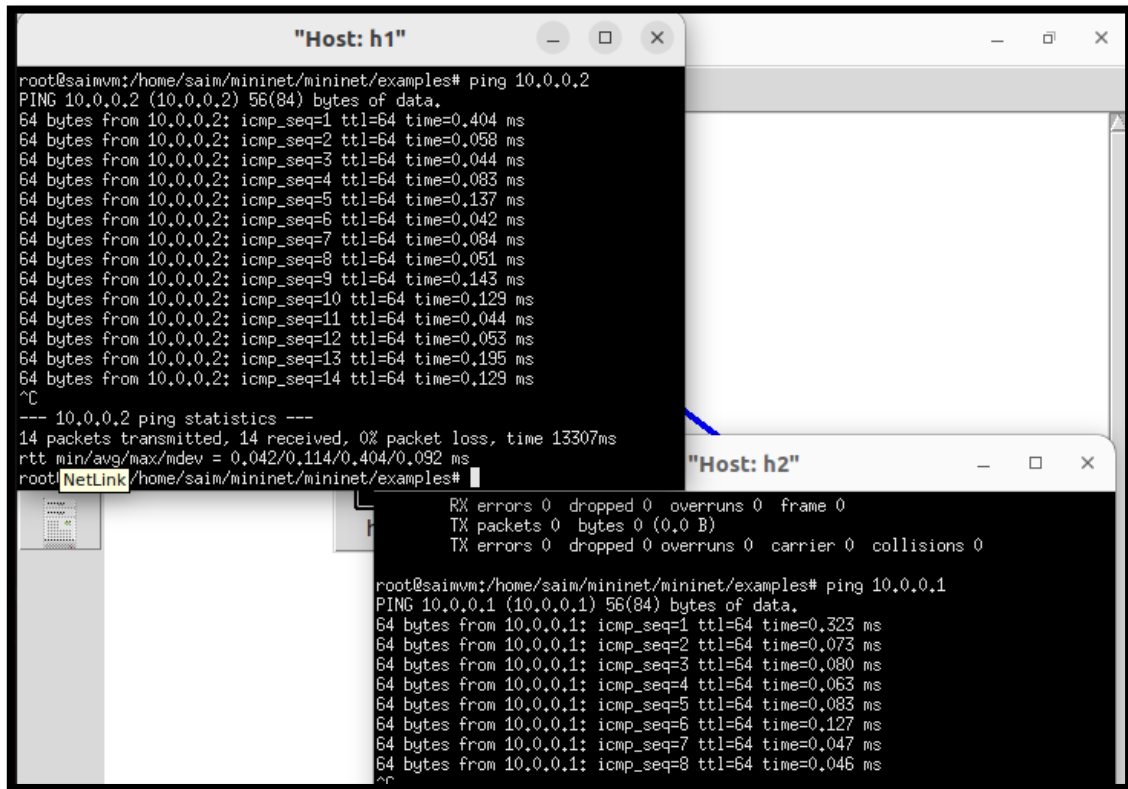
ifconfig



On host h1's terminal, type the command shown below. This command tests the connectivity between host h1 and host h2.

Ping 10.0.0.2

Do same on h2 and ping h1 IP address.



The screenshot shows two terminal windows. The top window, titled "Host: h1", displays the output of a ping command from 10.0.0.1 to 10.0.0.2. It shows 14 successful pings with varying response times, followed by statistics: 14 packets transmitted, 14 received, 0% packet loss, and a total time of 13307ms. The bottom window, titled "Host: h2", displays the output of a ping command from 10.0.0.2 to 10.0.0.1. It also shows 8 successful pings with varying response times, followed by statistics: 8 packets transmitted, 8 received, 0% packet loss, and a total time of 13307ms.

```

root@saimvm:/home/saim/mininet/mininet/examples# ping 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data:
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=0.404 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.058 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.044 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.083 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.137 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.042 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.084 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=0.051 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=0.143 ms
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=0.129 ms
64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=0.044 ms
64 bytes from 10.0.0.2: icmp_seq=12 ttl=64 time=0.053 ms
64 bytes from 10.0.0.2: icmp_seq=13 ttl=64 time=0.195 ms
64 bytes from 10.0.0.2: icmp_seq=14 ttl=64 time=0.129 ms
^C
--- 10.0.0.2 ping statistics ---
14 packets transmitted, 14 received, 0% packet loss, time 13307ms
rtt min/avg/max/mdev = 0.042/0.114/0.404/0.092 ms
root@NetLink:/home/saim/mininet/mininet/examples#

RX errors 0 dropped 0 overruns 0 frame 0
TX packets 0 bytes 0 (0.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@saimvm:/home/saim/mininet/mininet/examples# ping 10.0.0.1
PING 10.0.0.1 (10.0.0.1) 56(84) bytes of data:
64 bytes from 10.0.0.1: icmp_seq=1 ttl=64 time=0.323 ms
64 bytes from 10.0.0.1: icmp_seq=2 ttl=64 time=0.073 ms
64 bytes from 10.0.0.1: icmp_seq=3 ttl=64 time=0.080 ms
64 bytes from 10.0.0.1: icmp_seq=4 ttl=64 time=0.063 ms
64 bytes from 10.0.0.1: icmp_seq=5 ttl=64 time=0.083 ms
64 bytes from 10.0.0.1: icmp_seq=6 ttl=64 time=0.127 ms
64 bytes from 10.0.0.1: icmp_seq=7 ttl=64 time=0.047 ms
64 bytes from 10.0.0.1: icmp_seq=8 ttl=64 time=0.046 ms
^C

```

To stop the test, press Ctrl+c. The figure above shows a successful connectivity test. Host h1 (10.0.0.1) sent packets to host h2 (10.0.0.2) and successfully received the expected responses.

Stopping emulation:

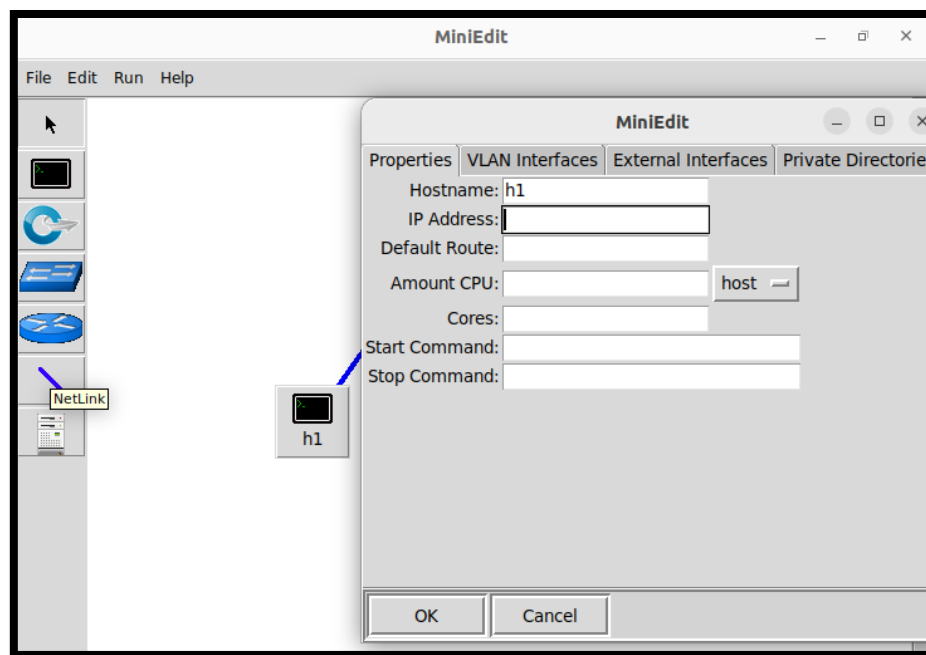
To stop the emulation, click on STOP button below the RUN button.



Automatic assignment of IP addresses

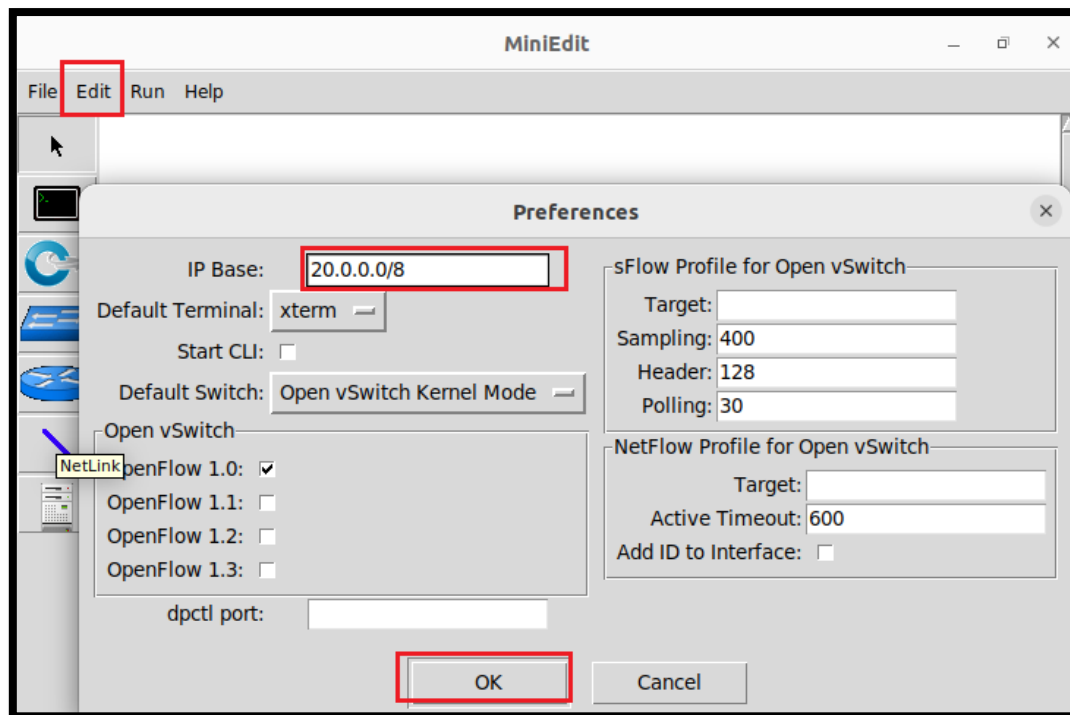
In the previous section, you manually assigned IP addresses to host h1 and host h2. An alternative is to rely on Mininet for an automatic assignment of IP addresses (by default, Mininet uses automatic assignment), which is described in this section.

Remove the manually assigned IP address from host h1. Hold right-click on host h1, Properties. Delete the IP address, leaving it unassigned, and press the OK button as shown below. Repeat the procedure on host h2.



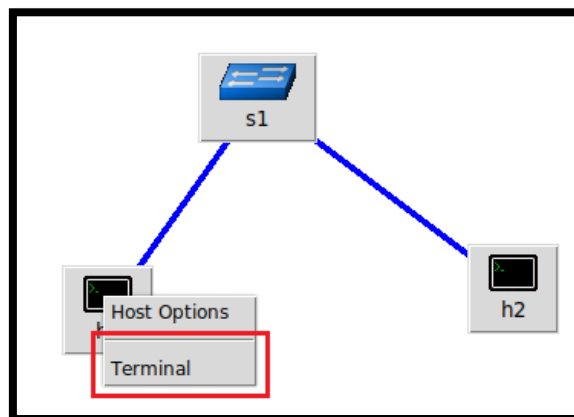
Do the same for h2.

Click on Edit, Preferences button. The default IP base is 10.0.0.0/8. Modify this value to 20.0.0.0/8, and then press the OK button.

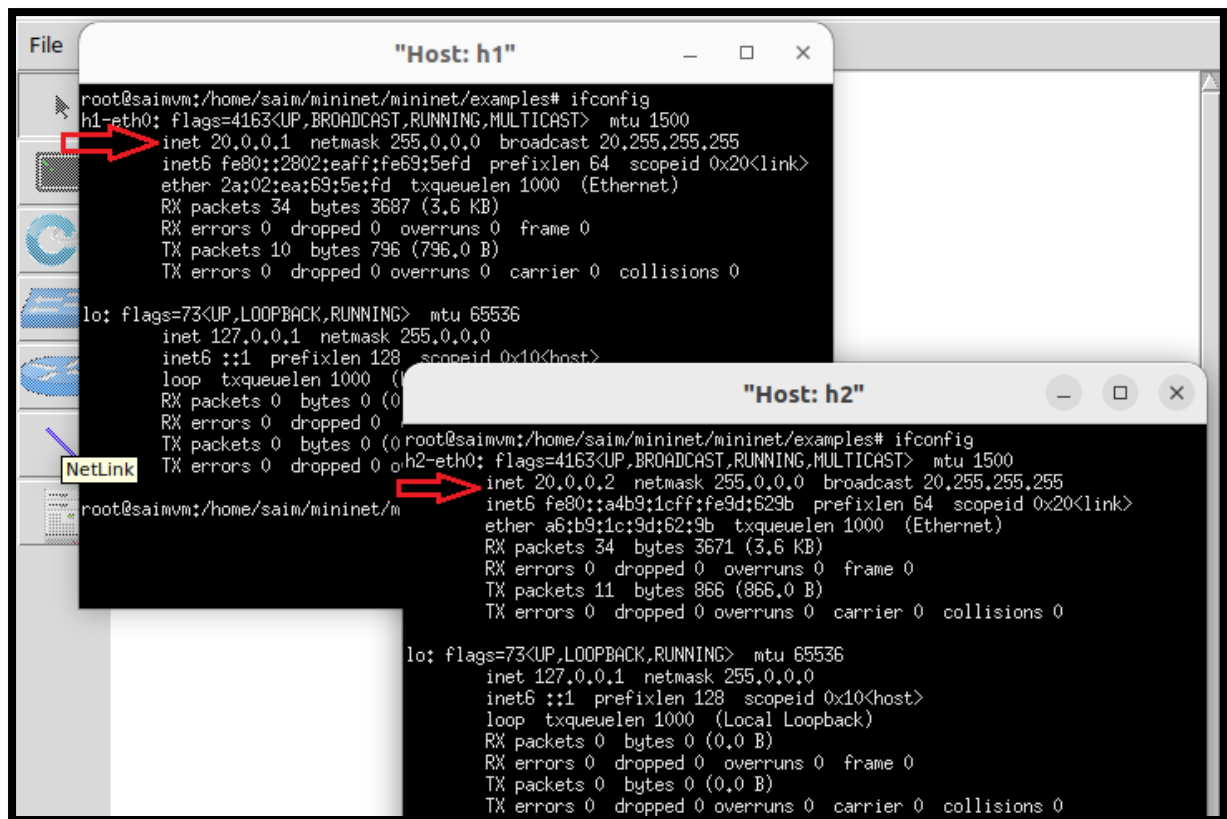


Run the emulation again by clicking on the Run button. The emulation will start and the buttons of the MiniEdit panel will be disabled.

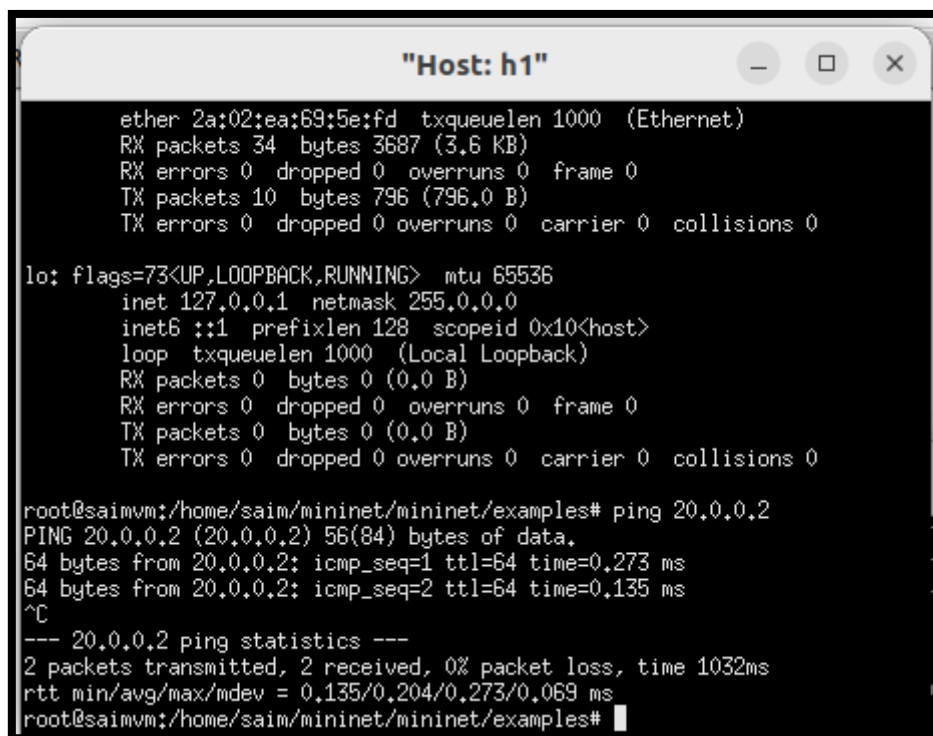
Open a terminal on host h1 by holding the right click on host h1 and selecting Terminal.



Type the command shown below to display the IP addresses assigned to host h1. The interface h1-eth0 at host h1 now has the IP address 20.0.0.1 and subnet mask 255.0.0.0.



Try ping h2 from h1 to check the connectivity.

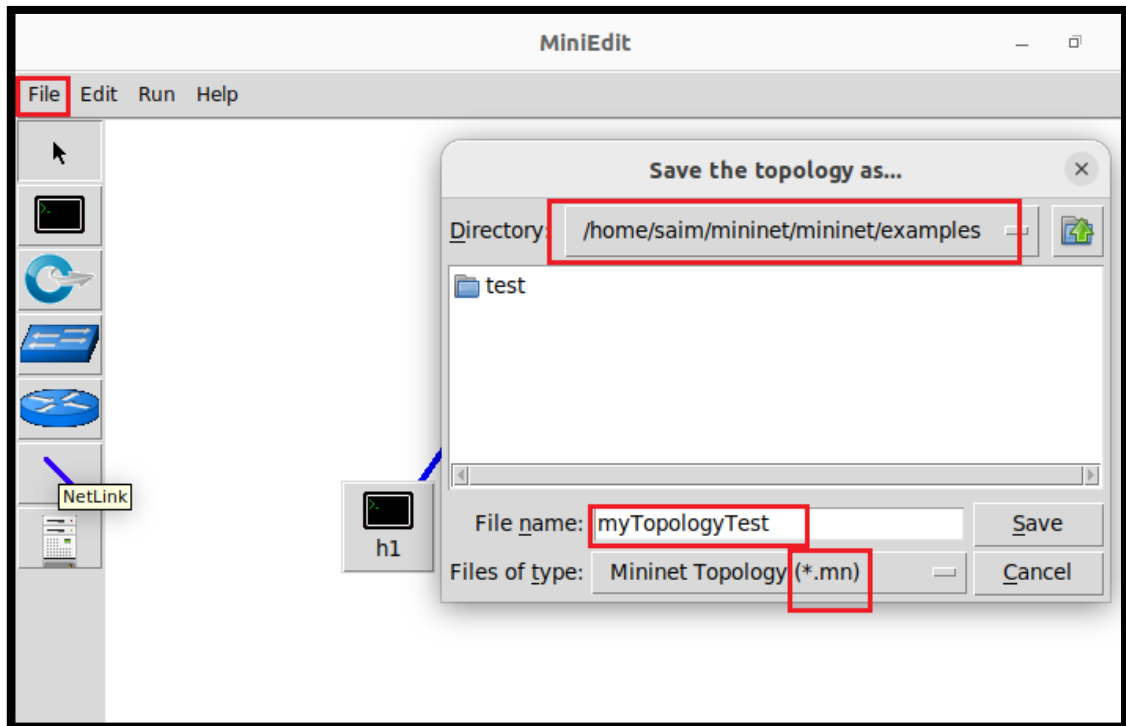


After this you can stop the emulation.

Save and load a Mininet topology

In this section you will save and load a Mininet topology. It is often useful to save the network topology, particularly when its complexity increases. MiniEdit enables you to save the topology to a file.

Step 1. Save the current topology by clicking on File then Save. Provide a name for the topology and save it in the local folder. In this case, we used myTopologyTest as the topology name.

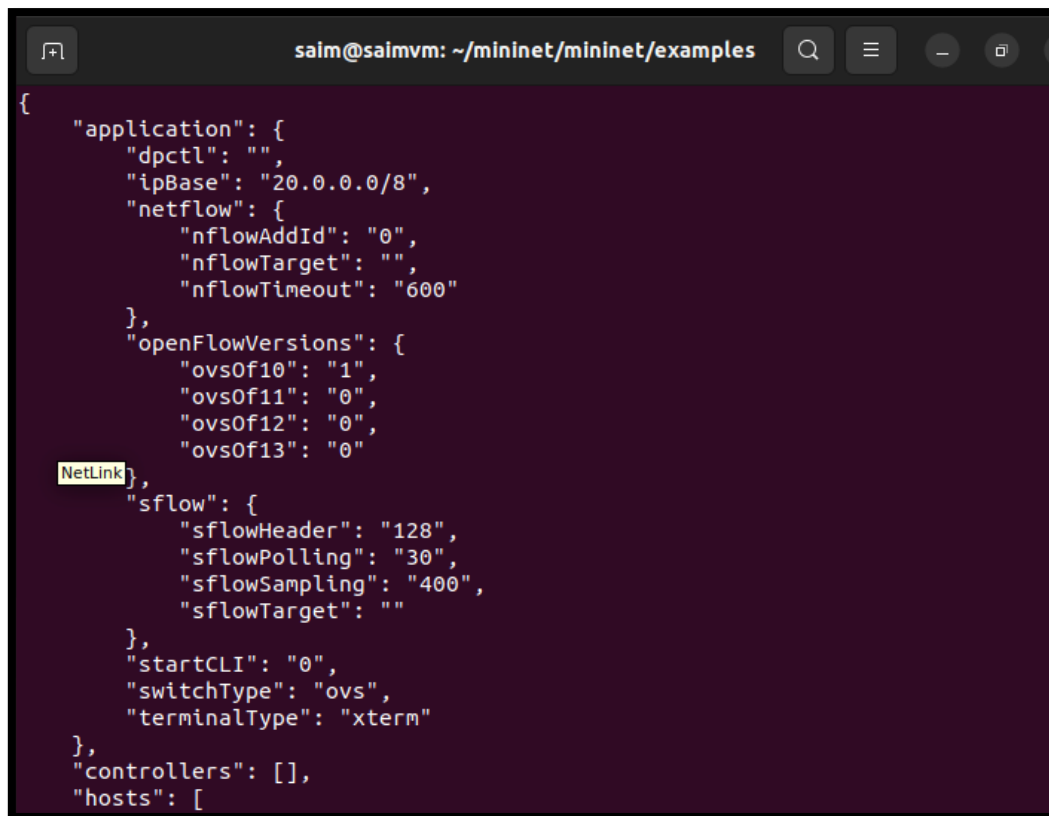


To load the topology, again go to the File and select OPEN to open the saved topology file. The files will be saved with .mn extension.

***The running and stopping of emulation can be seen in the Terminal of host machine as it will show all the activities.

You can also check the file contents by opening that saved topology file under examples folder, where it was saved. Use following command to open the file.

```
Sudo vi myTopologyTest.mn
```



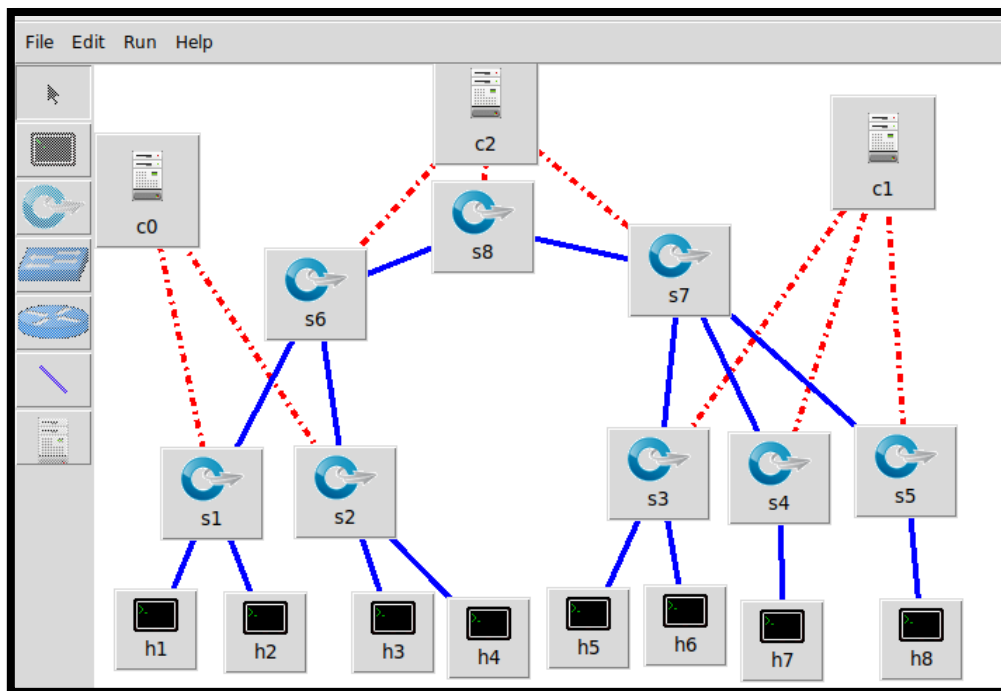
You can also save your topology in a python file by going to File menu and selecting 'Export to level 2 script'.

After exporting this .py file.

Run this file from mininet main console to see if the topology is created and running.

TASK:

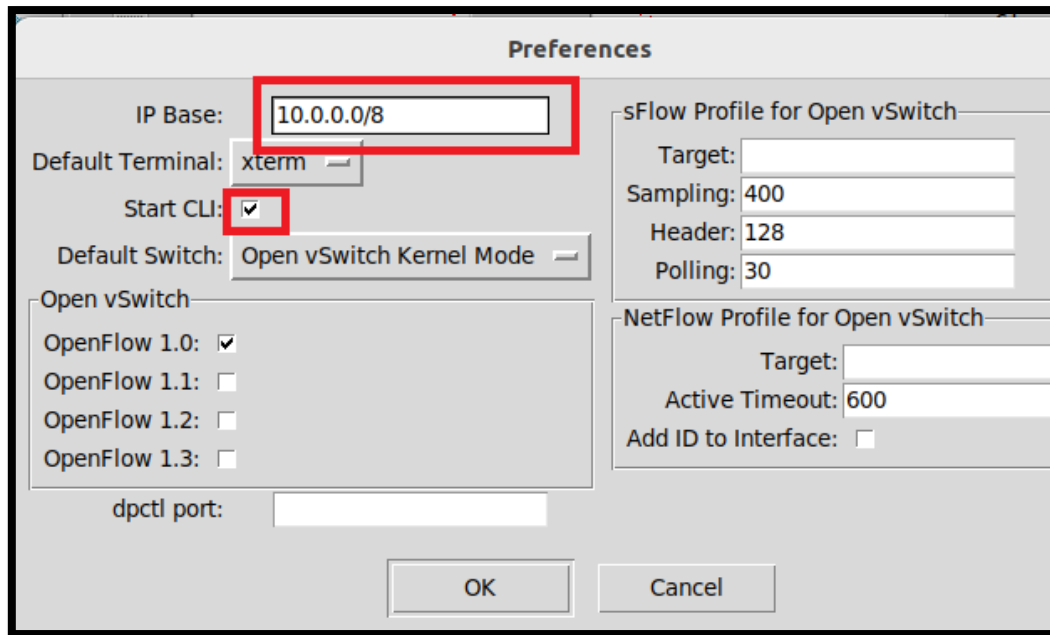
Create the topology as shown in the figure below. You can create this in a minute as you just need to select host and switch button (not the legacy switch).

**Controller configuration:**

Right click on Controllers and change their default port of 6633 to 6633, 6634, 6635 on C0, C1, and C2.

Setting preferences:

Now, go to preferences and select the CLI and make sure IP address is mentioned, which can automatically assign the IP address to all hosts.



Now we have a software-defined network scenario that should allow each host to communicate with any other host in the network.

We will save the MiniEdit topology file so we can load this scenario into MiniEdit in the future. We will also export a Mininet Python script that we can run in a terminal window to execute the scenario.

- To save the Mininet Topology (*.mn) file, click on File in the top menu bar and select Save from the drop-down menu. Type in a file name and save the file.
- To save the Mininet Custom Topology (*.py) file, click on File in the top menu bar and select Save Level 2 Script from the drop-down menu. Type in the file name and save the file.

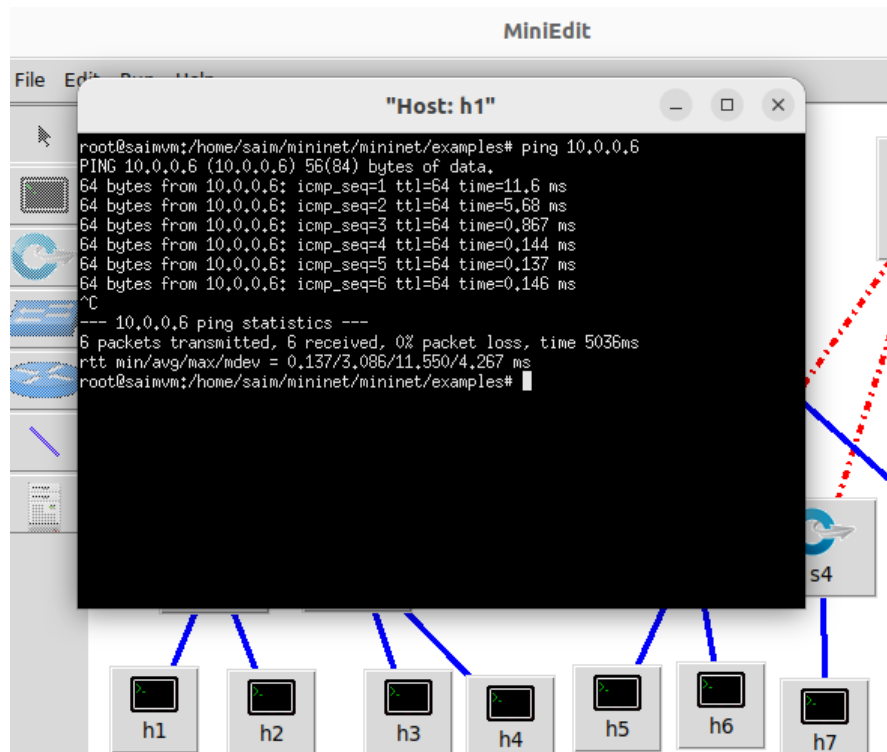
Run your emulation.

Click on RUN button.

Testing the configuration:

Right click on h1 and h6 and open their terminals. Run ifconfig command to check their IP addresses.

Now, ping h6 from h1 by typing the command ping 10.0.0.6



Because, you selected the CLI option in the preferences. You will see the changes and configuration in the main terminal window as follows.

```

Getting Links.
*** Configuring hosts
h1 h2 h3 h4 h5 h6 h7 h8
**** Starting 3 controllers
c0 c1 c2
**** Starting 8 switches
s1 s2 s3 s4 s5 s6 s7 s8
No NetFlow targets specified.
No sFlow targets specified.

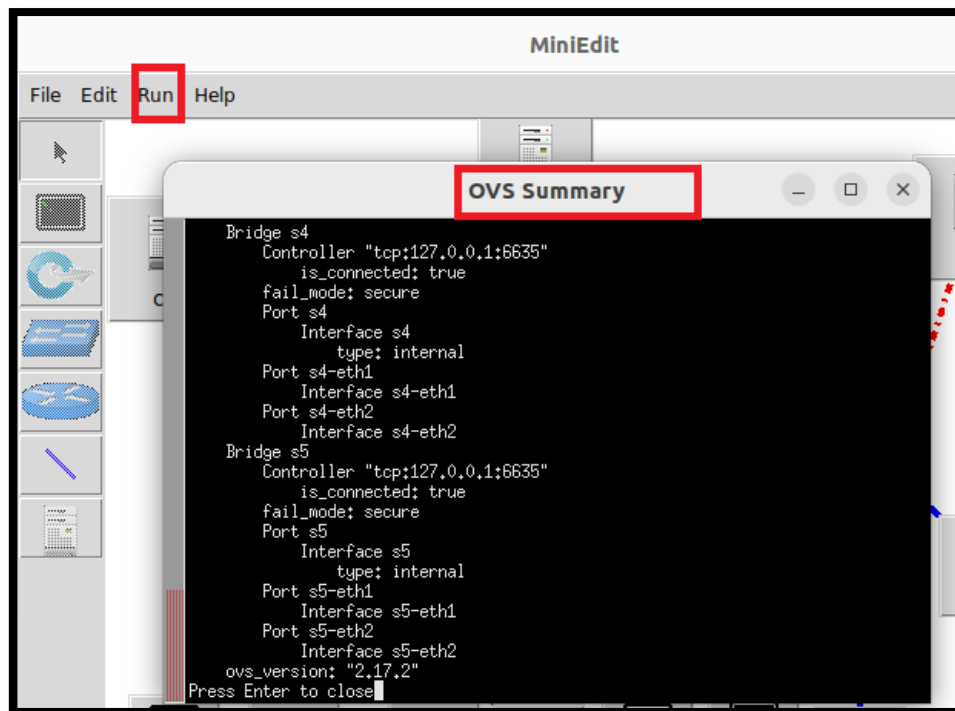
NOTE: PLEASE REMEMBER TO EXIT THE CLI BEFORE YOU PRESS THE STOP BUTTON. Not
iting will prevent MiniEdit from quitting and will prevent you from starting
e network again during this session.

*** Starting CLI:
mininet>

```

View Open vSwitch configurations

First, check the switch configurations in the network simulation to verify that everything is set up correctly. You can run the MiniEdit menu command, Run → Show OVS Summary to see an listing of switch configurations. In this case, we can verify that each switch is listening to the correct controller on the correct port.



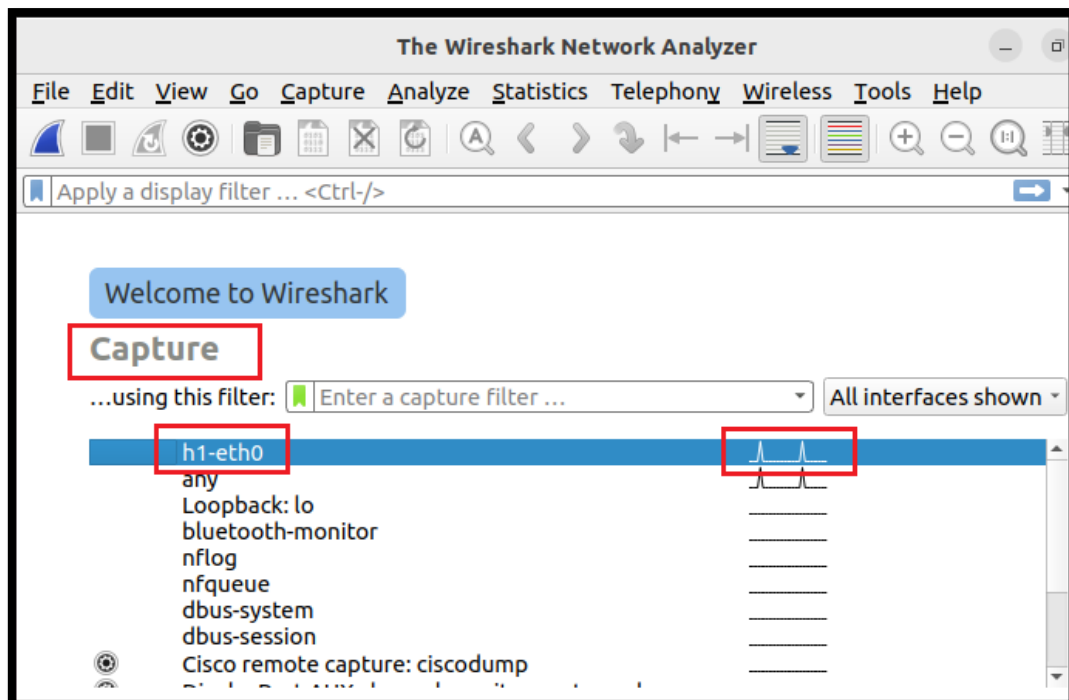
Traffic Analysis:

Wireshark

You can type the following command on h1 terminal and it will open the wireshark as shown below in the figure.

Wireshark &

Click on h1-eth0 and then click on capture.

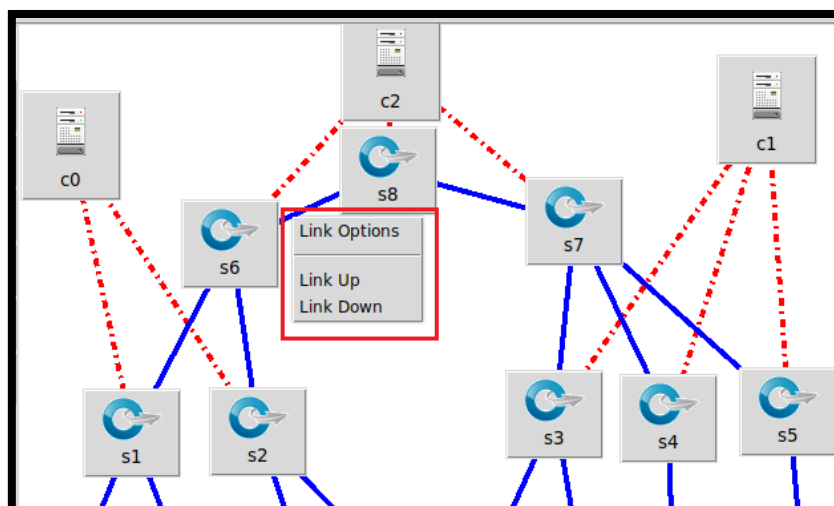


In parallel try to ping h1 from another host to see the traffic.

TCPDUMP:

You can also use the tcpdump command to capture traffic. Open h1 terminal and type tcpdump command mean while try to ping h1 from h6 or any other host machine.

Finally, you can also simulate a scenario by making a link down as follows. Right click on any link and it will show you link up and link down in the menu.



Finally, open the exported py file to see the code in python. Also, run this file in your host terminal to see how it works.